

BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration

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1. Introduction

What is EEG-based Motor Imagery?

Electroencephalography-based motor imagery (EEG-MI) is a technology that reads brain signals when a person imagines moving, allowing those signals to be translated into commands for electronic devices (Wolpaw et al., 2002). It can be used to help people control wheelchairs, robotic arms, and computer interfaces.

However, these brain signals vary from person to person and can also change over time (Lotte et al., 2018), so each MI session usually needs a calibration step before it can work properly.



Source: The University of Texas at Austin News

The Traditional Calibration Method

Calibration is the process where the system collects brain signals from the user while they perform specific tasks, then learns from those signals. The Graz method asks users to follow repeated on-screen prompts and imagine the same movement many times during each session. This method is still effective, but it usually takes 15–30 minutes per session, requires expert supervision, and can be tiring (Giles et al., 2022). As a result, it may reduce user engagement and lower data quality over time (Lotte et al., 2018).

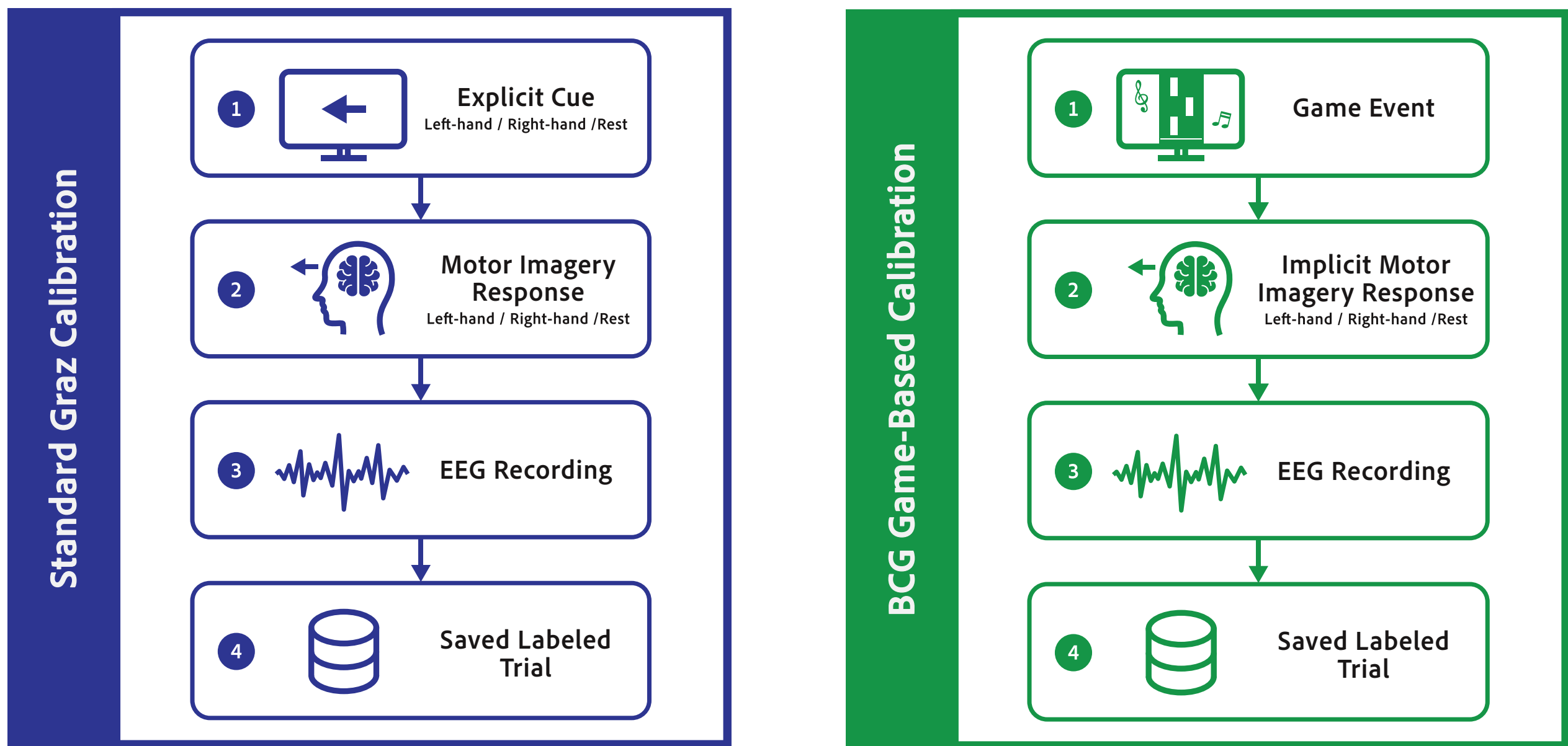
2. Objectives

- 1 Develop a dual-mode BCI software calibration framework (traditional Graz and gamified protocols)
- 2 Evaluate user experience by comparing fatigue and workload using NASA-TLX.
- 3 Establish a hardware-compatible baseline classifier for 14-channel motor imagery.
- 4 Build an automated pipeline for data ingestion and personalized fine-tuning.

3. Methodology

BCG Concept Compared to Graz Method

In the standard Graz protocol, users follow explicit on-screen cues to repeatedly imagine left-hand, right-hand, or rest states, producing clearly labeled EEG trials but often creating a repetitive and fatiguing experience. In contrast, BCG embeds the same calibration targets within gameplay events, so the user performs the required motor imagery tasks more naturally through interaction with the game. This design aims to preserve structured data collection while reducing boredom and mental strain, enabling labeled calibration data to be gathered implicitly rather than through direct commands.

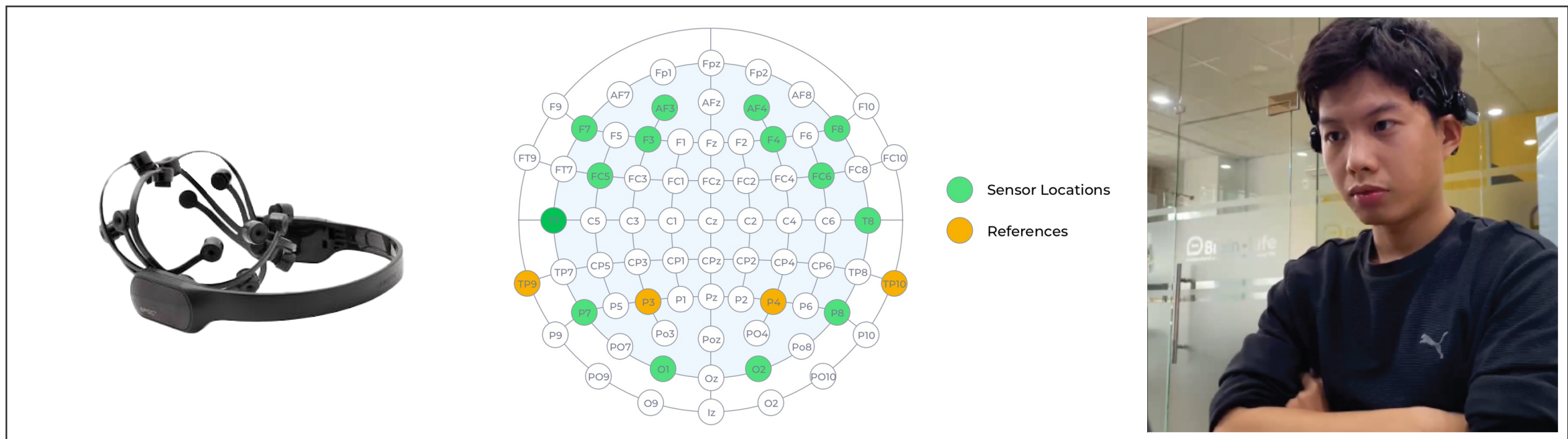


BCG Game Mechanic Design

When designing BCG, we framed it as an MI-as-data-collection game, where the primary goal is to collect correctly labeled EEG data for calibration rather than to enable free player control. The game offers only two choices at each decision point, corresponding to left-hand and right-hand motor imagery, while rest trials are recorded in the intervals between them. This ensures that each in-game event maps clearly to a motor imagery class and supports covert cueing instead of explicit instructions, as in the Graz protocol.

Device Selection and Setup

The experimental device used in this project is the Emotiv EPOC X headset. It provides 14 EEG channels placed according to the standard 10-20 system, including AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. It also supports real-time signal acquisition with wireless connectivity and contact quality monitoring. Emotiv's Cortex API is used to connect the headset to the computer and stream raw EEG data for recording and processing.



Emotiv EPOC X headset, electrode placement, and the author wearing the headset.

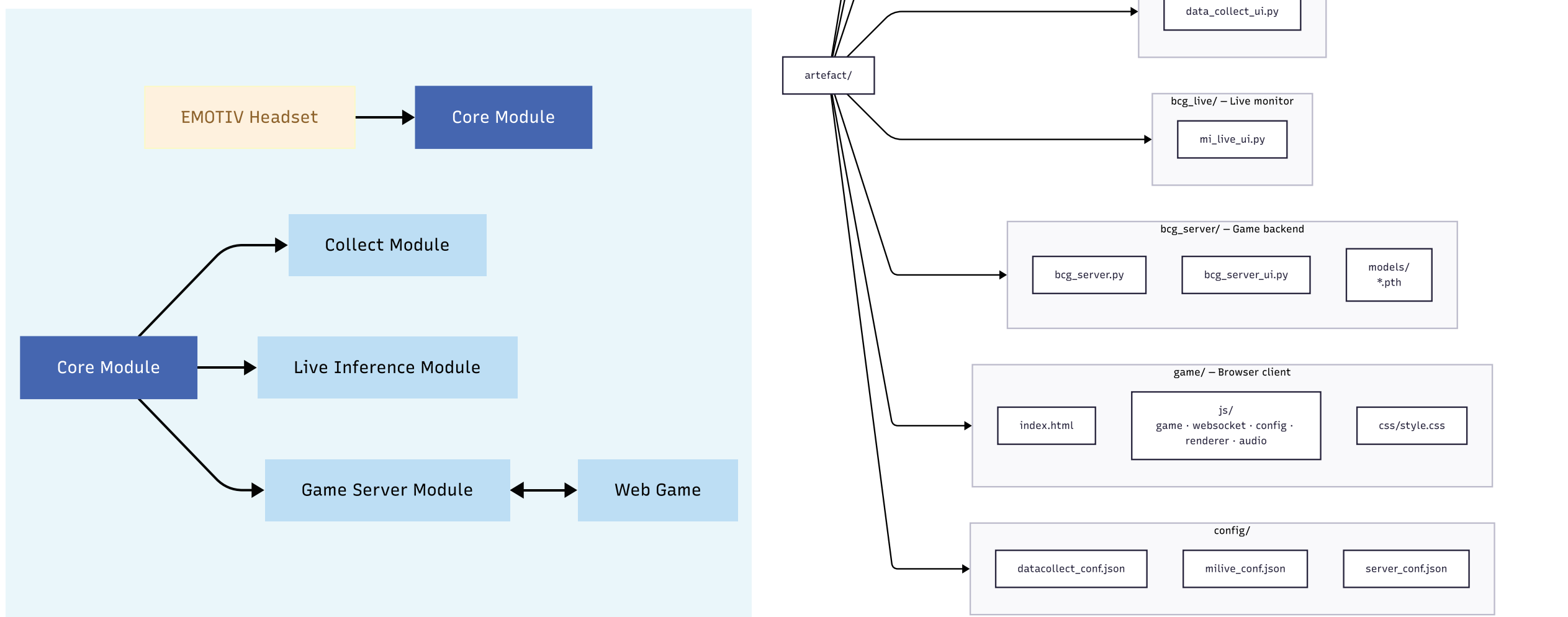
4. Artefact

System Overview

BCG includes a web game and a launcher with three main functions: Graz-based calibration, game server for gamified calibration, and live inference. In the gamified mode, the web game connects to the backend through WebSocket.

The system is divided into 5 modules: the core module, the collect module for the Graz method, the live inference module, the game server module, and the web game module. Shared config files and pretrained model weights are used across these modules.

The core module handles EEG acquisition from the EMOTIV headset, supports simulation mode, and performs signal preprocessing and classification. The Graz module provides the standard cue-based recording workflow and saves labeled trials to disk as .npz. The live inference module runs the pretrained classifier in a standalone sliding-window setting. The game server module manages trial collection during gameplay. The web game module implements the rhythm game and sends trial start/end markers over WebSocket to mark the EEG segment boundaries for each trial.



BCG overall system architecture and project structure

Baseline Dataset and Model

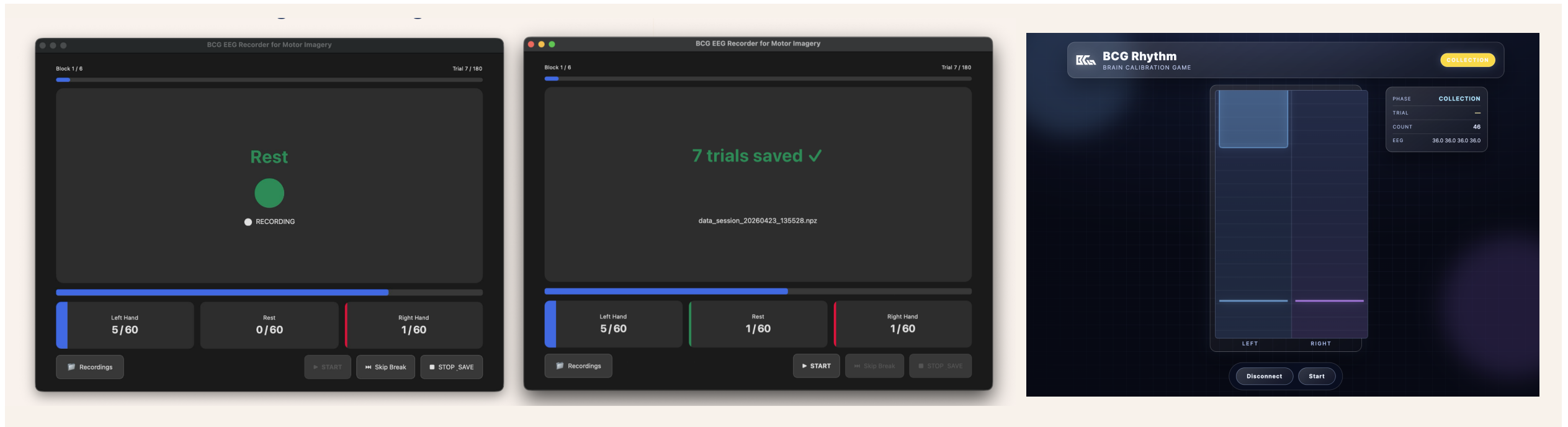
To establish a trial baseline for these two calibration modes, the project first explored a classical machine learning approach using Common Spatial Pattern (CSP) with Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) classifiers on the 14-channel MIMED dataset recorded with the EMOTIV EPOC X hardware. However, the initial results were not strong enough to serve as a reliable final baseline, mainly due to the limited size of the dataset and the difficulty of generalizing across subjects.

The final baseline was therefore built using EEGNet with transfer learning. The model was pretrained on BCI Competition IV 2a and then fine-tuned on MIMED to adapt it to the 14-channel hardware setup, and this configuration achieved the best performance at 72.08% accuracy in leave-one-trial-out evaluation.

User Interface

For the Graz interface, the system uses explicit on-screen cues to instruct users when to imagine left-hand, right-hand, or rest states, producing clearly labeled EEG trials but often creating a repetitive and fatiguing experience.

In contrast, the game-based interface embeds the same calibration targets within a two-lane tile-dropping gameplay, where the user performs the required motor imagery task when a tile on the corresponding lane reaches the target line. Each tile is also paired with a corresponding sound, and the notes gradually combine into a piano melody as the game progresses. This design preserves structured data collection while making the calibration process more interactive and engaging.



Graz-style motor imagery calibration interface vs. BCG gamified motor imagery calibration interface

5. User Study

User Study Design

Between-subjects user study with 10 healthy participants to compare the Graz protocol and the gamified protocol. Participants were split into two groups, with 5 people in each condition. Each participant completed a 10-minute calibration task, and we used NASA-TLX to measure perceived workload and user experience.

User Study Result

The gamified protocol reduced the overall workload compared with the Graz method. The clearest improvement was observed in Mental Demand, which showed a significant reduction in the gamified condition. Temporal Demand and Frustration also decreased, while Physical Demand, Performance, and Effort changed only slightly.

Despite the small sample size of n=5 participants per group, Welch's t-test confirmed a significant reduction in Mental Demand (p=0.047, d=1.55). This suggests that the gamified interface helped reduce the cognitive burden of the calibration task. The other NASA-TLX dimensions did not show significant differences between the two methods.

NASA-TLX Dimension	Graz	Gamified	p-value	Cohen's d
Mental Demand	7.8 ± 0.8	6.0 ± 1.4	0.047*	1.55
Physical Demand	2.6 ± 3.2	2.0 ± 0.7	0.702	0.26
Temporal Demand	4.0 ± 2.9	2.6 ± 1.1	0.362	0.63
Performance	3.8 ± 3.3	3.0 ± 3.5	0.720	0.23
Effort	4.0 ± 2.8	4.2 ± 2.0	0.902	-0.08
Frustration	4.2 ± 4.0	3.2 ± 1.6	0.628	0.33

Left: NASA-TLX score table with p-values from Welch's t-test and Cohen's d; Right: radar chart comparing Graz and Gamified average workload scores.

6. Conclusion

- 1 The project developed a dual-mode BCI calibration framework, including both the traditional Graz method and the gamified BCG method.
- 2 The gamified BCG protocol reduced perceived workload compared with the traditional Graz method, especially in Mental Demand, and assuming the same hardware setup, it can maintain comparable data quality while making calibration less mentally demanding.
- 3 Finding a dataset that matched the device setup was a major challenge during model training, and EEGNet was selected as the final baseline model after testing CSP+LDA, CSP+SVM, and EEGNet.
- 4 The self-finetuning calibration pipeline remains a work in progress.
- 5 Overall, the findings suggest that gamification can make motor imagery calibration more engaging and practical, with potential for wider real-world use in BCI systems.

References

Giles, J., Ang, K., Phua, K. and Arvaneh, M. (2022) 'A transfer learning algorithm to reduce brain-computer interface calibration time for long-term users', *Frontiers in Neuroergonomics*, 3, p. 837307.

Lotte, F., Bougrain, L., Cichocki, A., Clerc, M., Congedo, M., Rakotomamonjy, A. and Yger, F. (2018) 'A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update', *Journal of Neural Engineering*, 15(3), p. 031005.

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